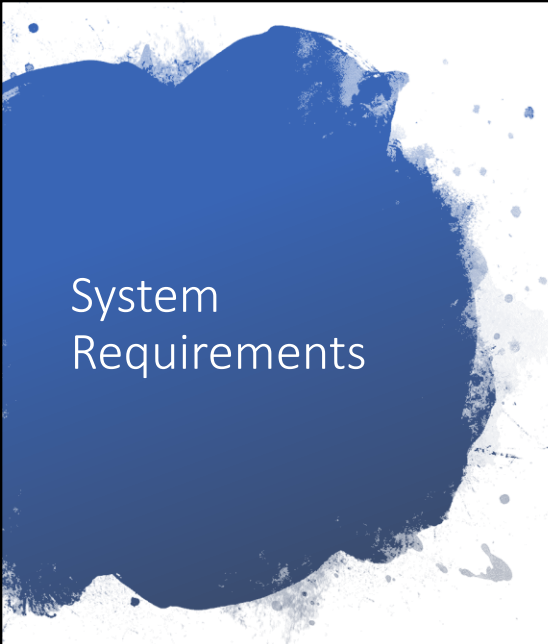


A graphic element consisting of two overlapping circles, one light blue and one dark blue, positioned behind the 'DriverPass' text.

DriverPass

System Analysis & Product Design



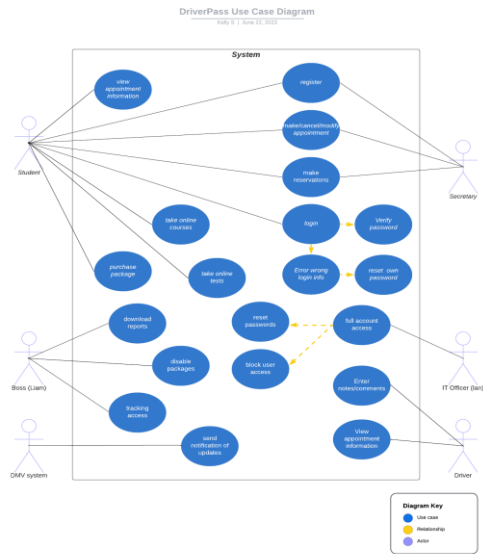
System Requirements

- Connected to DMV
- Validate credentials upon login
- Web-based
- Speeds capable of seamless video viewing

Two functional requirements considered for this system were connection to the DMV to facilitate updates to keep teaching information relevant and validating credentials upon login for security purposes.

Two non-functional requirements considered for this system were a web-based system to facilitate users connecting from anywhere with web access and speeds capable of seamless video viewing to facilitate the capability and good user experience of viewing video lessons.

Stakeholder Use Cases



For this system we have 6 entities interacting with the system. These entities are represented by the stick figures. They are:

- The Student
- The Secretary
- The Boss (Liam)
- The IT Officer (Ian)
- The DMV
- And the Driver

Each of these entities will have their own way of using the system. These uses are in the blue ovals and connected to the entity that will be using the system in that manner. If you see a yellow dotted arrow connecting uses it represents a relationship between them.

I will quickly go over this.

The Student will be able to:

- View appointment information
- Take online courses

- Take online tests
- Purchase packages
- Login

You will notice that login has a relationship with verification of that password or an error for a wrong password. If there is a wrong password the student will have an option to reset their own password.

The student and the secretary will both be able to:

- Register
- Make, cancel, or modify an appointment
- And make reservations

The Boss will have the ability to:

- Download reports
- Disable packages
- And have access to tracking information

The IT Officer will have full account access which will allow them to reset passwords and block user access.

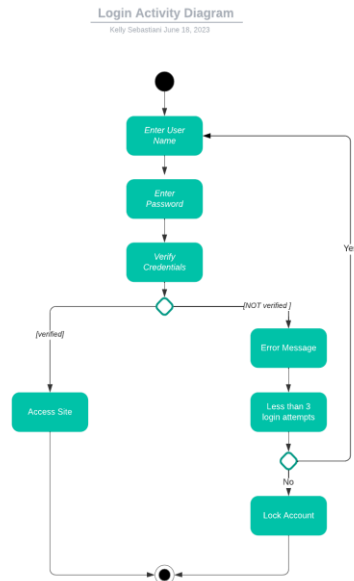
The DMV system will be connected to send notifications of any updates.

And lastly the drivers will be able to:

- Enter notes or comments about their lessons
- And can view appointment information

You will notice that this system was tailored to the wants and needs that were discussed in prior meetings.

User Authentication Workflow



This diagram depicts the activity of logging in. Starting from the top the user will:

- Enter their username and
- Enter their password

These credentials will then be verified. If verified the user will gain access to the site and the activity ends. If these credentials are not verified the user will receive an error message. If the user has less than 3 failed attempts of logging in, they will be able to try again. If not, the account will be locked for security purposes.



Security Considerations

- Account Locks after multiple failed login attempts
- Password requirements to enforce strong password choices
- Verification of credentials
- User specific access
- System activity tracking
- Keeping user sensitive data private
- Allowing students and drivers to view lesson information

For security data and personal safety were both considered. To aid in data security some features we incorporated are...

- Accounts locking
- Password requirements
- Verification of credentials
- User specific access
- System activity tracking
- And keeping personal information private

For personal security students and drivers will be able to view the lesson information prior to meeting. Tracking will also play a role in protecting the students and drivers.



Product Limitations

- Cannot modify or update information offline
- Cannot add or remove packages

Limitations for the system include the inability to modify or update information offline and the inability to add or remove packages.

The inability to modify or update information may be inconvenient but does protect from duplicate data being created.

The inability to add or remove packages is something that may be able to be implemented in the future, but for now the system will have the capability of disabling packages when needed.

Key Takeaways

- Requirements were translated into a complete system design.
- Security and usability were incorporated into the solution.
- UML artifacts documented workflows, interactions, and architecture.
- The proposed design supports future scalability and maintainability.